

Title:

No association between genetically predicted C-reactive protein levels and colorectal cancer survival: a Two-sample Mendelian randomization analysis

Running title: CRP₇ and colorectal cancer survival

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1 Abstract

2 Objectives: Elevated C-reactive protein (CRP) ~~is levels are~~ associated with an increased risk for
3 ~~colorectal cancer (CRC, as well as a poor prognosis, and poor prognosis of, colorectal cancer (CRC),~~
4 but it remains unclear whether these associations are causal. This study examined ~~the~~ potential causality
5 between CRP levels and CRC survival using ~~two~~2-sample Mendelian randomization (MR).

6 Methods: From the Korean Genome and Epidemiology Study, a genome-wide association study (n=
7 59,605), 7 single-nucleotide polymorphisms (SNPs) related to ~~log~~2-transformed CRP levels were
8 extracted as instrumental variables for CRP levels. The associations between the genetically predicted
9 CRP and CRC-specific and overall mortality among CRC patients (n=6,460) were evaluated by Aalen's
10 additive hazard model. The sensitivity analysis excluded ~~a the~~ SNP related to the blood lipid profile.

11 Results: During a median of 8.5 years of follow-up, among 6,460 CRC patients, ~~2,676 (41.4%) CRC~~
12 ~~patients died, from all causes and 1,622 (25.1%) died from CRC.~~ Genetically predicted CRP ~~was~~
13 ~~levels were~~ not significantly associated with ~~the~~ overall or CRC-specific mortality in CRC patients. The
14 hazard difference per 1,000 person-years ~~(95% confidence interval)~~ for overall and CRC-specific
15 mortality per ~~two~~2-fold increase in CRP ~~levels~~ was -2.92 (95% CI, -14.05–to 8.21) and -0.76 (95% CI,
16 -9.61–to 8.08), respectively. These associations were consistent in ~~a~~ subgroup analysis according to
17 metastasis and ~~a~~ sensitivity analysis excluding ~~the~~ possible pleiotropic SNPs.

18 Conclusions: Our findings do not support ~~a~~ causal roles for ~~genetically predisposed~~ CRP levels in CRC
19 survival.

20

21 Keywords:

22 Colorectal neoplasm, C-Reactive Protein, Survival Analysis, Mendelian Randomization
23 Analysis

서식 있음: 아래 첨자

메모 포함[오전1]: Is this what you meant—i.e., does this refer to a single SNP? This seems to be the case based on the main text. If not, this could be changed to "SNPs". If a single SNP is referred to, "a" is appropriate here because that SNP was not discussed earlier in the abstract.

메모 포함[오전2]: The original wording here was not a grammatical mistake, so if you strongly prefer the original, it would be possible to revise back, but this wording is suggested as a clearer option that is easy to interpret.

INTRODUCTION

C-reactive protein (CRP) is a widely used as a clinical parameter of to measure inflammatory status, and numerous observational studies have reported associations between elevated CRP levels and the risk of colorectal cancer (CRC) incidence risk [1,2], population mortality due to CRC [3,4], and the clinical prognosis of colorectal cancer (CRC) patients, including death [5,6], metastasis [5,7], postoperative complications [8,9], and recurrence [5,10,11]. For the prognosis of CRC, three-Three studies of systematic reviews and meta-analyses have updated-provided up-to-date information on the association between elevated CRP levels and a poor prognosis of CRC [10,12,13]. For various other solid tumors, as well as CRC, CRP is is one of the variables that must be included when constructing a prognostic prediction model [10], including the Glasgow Prognostic Score (GPS), which is known to effectively predict the prognosis of CRC patients [14]. However, in these observational studies, it is difficult to differentiate between the deterioration of cancer-related inflammation and the clinical impact of elevated CRP levels themselves, because most studies used the analyzed peri-treatment CRP levels. The best timing for the evaluation of evaluating CRP-related markers is still remains unclear [6]. In addition, since the association between cancer progression and an immune response is often bidirectional and multifactorial, it is difficult to avoid reverse causality and residual confounding effects in observational studies [15]. To overcome these limitations and clarify the true causality, Mendelian randomization (MR) using genotypes as instrumental variables has been widely applied. Because genetic variants are randomly allocated by Mendel's law, an MR study using genetic variants as instrumental variables can be independent of potential confounders and can exclude the possibility of reverse causality. However, constructing large data sets with intermediate phenotypes and genetic instruments is challenging due to the high cost of measurements and/or the lack of suitable biological specimens. On the other hand, In this context, two-sample MR can evaluate the association between exposure and outcome using two-2 independent existing genome-wide association studies (GWASs), and the recent proportion of two-sample MR is steadily becoming more common increasing in studies in research using MR analysis [16]

GWASs have reported that several single-nucleotide polymorphisms (SNPs) are associated with CRP levels [17–19]; in the largest recent GWAS, those SNPs explained about 7.0% of the variance in CRP levels [17]. However, limited epidemiological data have been reported on the association between CRP-related genetic variants and the prognosis of CRC, and the findings are still inconsistent [20–23].

Recently, the The International Survival Analysis in Colorectal Cancer Consortium recently reported that genetically predicted CRP levels were not significantly associated with CRC-specific mortality in a GWAS study of 16,918 European CRC cases [24]. However, since CRP levels have an ethnic variation among different ethnicities [25] and several SNPs were found to be related to CRP levels only in a GWAS study of East Asians [19,26,27], there remains a need for further research to clarify the association between CRP levels and survival of CRC in other ancestries. Therefore, we evaluated the causal role of genetically predicted CRP levels in the survival of CRC in Koreans using by conducting a two-sample MR study, consisting of with Korean-representative Korean GWAS datasets.

METHODS

Sources of the CRP GWAS data

Supplementary Figure 1 shows the flow chart of the CRP GWAS. The association between SNPs and CRP levels was determined using the GWAS dataset from the Korean Genome and Epidemiology Study (KoGES) [28], a consortium project consisting of six prospective cohort studies supported by government funding. Over 223,000 participants have been recruited, with 72,298 from population-based studies (58,700 from the KoGES Health Examinee [KoGES_HEXA] study, 8,105 from the KoGES Cardiovascular Association study [KoGES_CAVAS], and 5,493 from the KoGES Ansan and Ansung Study) who provided epidemiological information and genome-wide arrays after the quality control procedure. The KoGES_CAVAS and KoGES Ansan and Ansung Study consisted of community inhabitants, while the KoGES_HEXA study consisted of participants recruited from the national health examinee registry. Of those From the original

sample, 10,358 participants whose serum CRP level was not measured, 146 participants with a serum CRP level ≥ 10 mg/L, 2,462 participants with a previous history of cancer, and 81 participants with missing values in-for a previous history of cancer were excluded from the analysis. Consequently, 59,605 participants were included in the final analysis (Supplementary Table 1). Genomic DNA was extracted from peripheral blood, and the GWAS was ~~done-conducted~~ using the Korean Biobank Array (K-CHIP) customized for the Korean population. Details on genotyping, GWAS quality control, and imputation have been described elsewhere [29]. K-CHIP contains 833,535 SNPs, including 89,413 SNPs present in East Asians. Imputation was conducted using the 1000 Genomes Phase 3 dataset of the East Asian population as a reference panel. Informed consent was obtained from the participants (Institutional Review Board [IRB] approval by the Korea Centers for Disease Control and Prevention in Korea; no. 2015-08EXP-01-C-A and 2016-02-20-C-A).

Source of the CRC GWAS data

The CRC GWAS data were obtained from the Hwasun Cancer Epidemiology Study-Colon and Rectum Cancer (HCES-CRC). ~~The HCES is a hospital-based case-control study aiming to identify the serologic and genetic risk factors for multiple cancers, including esophageal [30], breast [31], gastric [32], and colorectal cancers [33].~~ The HCES-CRC consisted of 7,089 hospital-based CRC cases and 4,979 population-based cancer-free controls. Details of genotyping and GWAS quality control have been described elsewhere [33]. ~~The baseline characteristics of the CRC GWAS are presents-presented in Supplementary Table 2.~~ In brief, the subjects were patients diagnosed with histologically confirmed CRC at Chonnam National University Hwasun Hospital between 2004 and 2014. Germline DNA genotyping was performed using the Infinium OncoArray-500K BeadChip (Oncoarray, Illumina, San Diego, CA, USA) in 3,158 CRC cases, and the Infinium Multi-Ethnic Global BeadChip (MEGA, Illumina, San Diego, CA, USA) in 3,465 cases. Of those, 163 cases without information on the TNM stage were excluded, ~~from the analysis,~~ and ~~finally~~ 6,460 CRC cases were finally included in ~~the analyses~~analysis. The cause and date of death were obtained from the National Statistical Office. The date of death was ascertained until December 31, 2020. The cause of death was coded according to the

International Classification of Diseases, 10th revision [2]. The details of the HCES-CRC and imputation procedure were have been described previously [34]. The analysis included SNPs with an info score greater than 0.4. All patients and controls gave informed consent to study participation at the time of peripheral blood collection. The HCES-CRC was reviewed and approved by the CNUHH IRB (CNUHH-2020-063).

Associations between genetic variants and CRP levels, and CRC survival

The This study consisted of the a discovery cohort of 47,258 individuals from KoGES_HEXa and the a replication cohort of 12,347 individuals from KoGES_CAVAS and the KoGES _Ansan and Ansung Study. Multivariate linear regression was performed to evaluate the association between genetic variants and log₂-transformed serum CRP levels. Age, sex, survey year, and the assessment centers of cohort studies were adjusted. The first 10 principal components were also adjusted to correct for the possible population structure in the GWAS. The statistical analyses were performed using PLINK (ver. 1.90b6.0). SNPs with a minor allele frequency (MAF) < 0.05 were excluded from the analysis, leaving 1,859 SNPs significantly associated with the log₂-transformed serum CRP levels in KoGES ($P < 5 \times 10^{-8}$). We used a linkage disequilibrium (LD)-based clumping cut-off of $r^2 < 0.001$ and a window size of 10,000 kb directly from the KoGES genotyping data. Results The results of the discovery and replication phases were are presented in Supplementary Table 3. Discovery analysis identified 13 significant SNPs associated with serum CRP levels, of which 7 were replicated. The replicated SNPs were rs2794520 near *CRP*, rs12133641 in *IL6R*, rs71086917 in *LINC02819*, rs1260326 in *GCKR*, rs7383869 near *IL6*, rs79320731 in *HNF1A*, and rs429358 in *APOE*.

The association between selected SNPs and log₂-transformed serum CRP levels was re-evaluated in a pooled analysis, and these results were used for the MR study. The minimum value of the *F* statistics of the selected SNPs was 61.2, and it is was expected that the bias by weak instruments in the main analysis will would not be significant.

Because the Aalen's additive hazard model preserves linearity, it can be used in two-sample MR analysis regardless of the proportional hazard assumption [35]. So Therefore, we conducted the

Aalen additive hazard regression to evaluate the association between CRP-related SNPs and CRC survival using the R package “timereg.”. Statistical analyses were performed using R version 4.2.0.

Two-sample Mendelian randomization and genetic risk score

We estimated genetically predicted CRP levels and CRC survival using the inverse-variance weighted (IVW) method using the R package “MendelianRandomization” [36].

The estimated associations of genetically predicted CRP levels with hazard differences (HDs) of-in mortality were expressed with respect to a two-fold increase in the serum CRP level.

Seven selected SNPs were selected as instrumental variables to calculate the weighted genetic risk score (GRS) for the log₂-transformed serum CRP level. Supplementary Figure 2 showed-shows the distribution of the weighted GRS for CRP and CRC GWAS. In PLINK, the weights are the estimated beta coefficients associated with each copy of the minor allele in a linear regression analysis, and the mean GRS per non-missing genetic marker was calculated and the GRS was divided into quintiles (Q1 to Q5).

Statistical power

To the best of our knowledge, there is no available tool to estimate statistical power for survival outcomes in Mendelian randomization. Instead, we used a conservative tool that considered binary survival outcomes as binary outcomes [37]. Of a total of 6,460 CRC cases, 2,676 (41.4%) deaths occurred over an 8-year follow-up period. In previous meta-analysis of CRC patients, the hazard ratios (HRs) for overall survival of elevated CRP levels and CRP-to-albumin ratio were 2.04 [12] and 2.03 [13], respectively. We have-had more than 90% power to detect an odds ratio (-OR) of 1.50 for the association between CRP levels and overall mortality at a significance level of 0.05, assuming that the GRS would explain 4.0% of the variance of-in CRP explained by the genetic variance levels.

In addition, we ran a simulation using the-an additive hazards model for power calculation. With 6,460-the number-of CRC cases and 2,676 deaths accrued over an-a 8-year follow-up, the population-averaged hazard was estimated to be $2,676/(6,460 \times 8) = 0.052$ per person*-year. We have

had at least 90% power to detect a 50.0% difference in hazard ($HD = 0.026$) for every 1 standard deviation (SD) increase of in $\log_2$ -transformed CRP levels, assuming that 4.0% of the variance of CRP was explained by the GRS. The R code for the simulation was conducted by modifying modified from the R code in the study of Hua et al. [24]

Sensitivity analyses

~~Mendelian randomization~~MR relied on ~~three~~3 assumptions. First, genetic variants are associated with the exposure (CRP). Second, genetic variants are not associated with potential confounders. Third, genetic variants are not directly associated with the outcome (death), except through the exposure (a lack of horizontal pleiotropic effects). In our study, the selected SNPs were validated in the CRP GWAS and were independent of each other (not in linkage disequilibrium LD). In addition, using Phenoscanner, a database of GWAS results, the pleiotropic effects of the selected SNPs was confirmed were checked [38], and the an association of rs429358 near APOE with the blood lipid profile was reported [39,40]. SoTherefore, we performed an additional sensitivity analysis excluding rs429358.

For ~~two~~2-sample MR, ~~three~~3 sensitivity analyses (simple and weighted median, and MR-Egger regression) were performed. Although the IVW method is sensitive to violations of the assumption on regarding pleiotropy, the results from the simple and weighted median are consistent even when up to 50% of the information comes from invalid instrumental variables [41]. The intercept estimated from the MR-Egger regression provides an estimate of the horizontal pleiotropic effect [42].

RESULTS

As shown in Table 1, showed that SNPs in 7 loci identified to be were associated with the $\log_2$ -transformed CRP levels, as follows: were rs2794520 near CRP ($P = 7.23 \times 10^{-200}$), rs12133641 in IL6R ($P = 3.52 \times 10^{-47}$), rs71086917 in LINC02819 ($P = 6.54 \times 10^{-16}$), rs1260326 in GCKR ($P = 1.31 \times 10^{-29}$), rs7383869 near IL6 ($P = 3.53 \times 10^{-37}$), rs79320731 in HNF1A ($P = 1.22 \times 10^{-99}$), and rs429358 in APOE ($P = 3.53 \times 10^{-125}$). The estimated $\log_2$ -transformed CRP variance explained by the selected SNPs was 4.0%.

185 Among these SNPs, rs2794520 (*CRP*) [17], rs12133641 (*IL6R*) [27,43], rs1260326 (*GCKR*)
186 [39], and rs429358 (*APOE*) [17] have been reported to be associated with CRP levels in previous studies.
187 For rs7383869 (*IL6*) and rs79320731 (*HNF1A*), the LD blocks of those SNPs contained rs2097677 (R^2
188 $=-0.42$) and rs7310409 ($R^2=-0.90$), respectively, as reported by the a previous GWAS of in the Japanese
189 population [26].

190 There are two-2 possible explanations for the association between the rest-remaining 1-novel
191 SNP, rs71086917 (*LINC02819*), and CRP levels. First, although, to our knowledge, the-an association
192 between *LINC02819* and CRP was-has not been reported in previous studies, rs10908724 in *LINC02819*
193 and in the LD block for rs71086917 ($R^2=-0.180$) was related to MCP-1 [44], which mediated-mediate
194 the chemotaxis of CRP [45]. However, since the results in-of the GWAS do not reveal the function of
195 the SNP, the association between the regulation of *LINC02819* and CRP or MCP-1 needed-still needs to be
196 evaluated. Second, rs10908724 may be a proxy SNP for rs3093068 (*CRP*) identified in a previous
197 GWAS [43]. In our genotype data, rs10908724 were-was correlated to rs3093068 ($R^2=-0.02$).

198 Table 2 presents the effects of a-genetically predicted CRP levels on the risks of death using
199 summary statistics and the GRS. The effects estimated by the IVW method shows-showed that a two2-
200 fold increase in serum CRP levels was not significantly associated with the risk of overall -and/or CRC-
201 specific mortality (HD per 1,000 person-years: -2.92 and -0.76, respectively; 95% CI: -14.05 to 8.21
202 and -9.61 to -8.08, respectively). And-theFurthermore, the results of sensitivity analyses using the
203 simple median and median weighted estimation are-were consistent with the main results. The MR-
204 Egger intercept showed no significant evidence of pleiotropic effects (p -value $P=0.669$ for overall mortality
205 and p -value $P=0.876$ for CRC-specific mortality). These non-significant results were similar in the sensitivity
206 analysis that excluded the SNP related to the blood lipid profile. The scatter plot of SNP-specific
207 associations with CRC-specific survival against coefficients of SNP-CRP associations and the
208 regression line depicting the association between genetically predicted CRP levels and survival was-are
209 visualized in Supplementary Figures 3 and 4, respectively. The linear association between a 1-SD
210 increment of the GRS for CRP levels was not significantly associated with CRC-specific mortality (HD
211 per 1,000 person-years (-2.09; 95% CI, -4.26 to -0.08). The GRS for CRP levels demonstrated a

메모 포함[오전3]: This seems like a fairly small R^2 value, although of course the interpretation of R^2 values depends on the context of the research question. Nonetheless, would it be possible to double-check that the number is correct and/or whether this should be phrased as “weakly correlated”?

significant association with overall mortality (HR per 1,000 person-years: -2.09; 95% CI: -5.77 to 0.59). Additionally, compared to the third quintile of the GRS for CRP levels, the HR for the first quintile of the GRS for CRP levels was 8.53 (95% CI: 1.12 to 15.94). However, these associations were not observed in a sensitivity analysis that excluded rs429358, a variant known to be associated with the blood lipid profile.

Table 3 presents the subgroup analysis according to metastasis. In the analyses using summary statistics and the individual GRS, the association between genetically predicted CRP levels and mortality were not significant, regardless of metastasis.

Discussion

This study did not find evidence for an association between genetically elevated CRP levels and survival among CRC patients in Korean. Consistent results were found in a sensitivity analysis excluding a possible pleiotropic SNP and a subgroup analysis according to metastasis.

The effect of CRP levels on survival was recently evaluated in CRC cases of European ancestry using two-sample MR [24]. Similar to our results, Hua et al. reported that genetically predicted CRP levels were not associated with CRC-specific mortality regardless of metastasis. Compared to the instrumental variants used in the study of Hua et al. [24], among the 7 CRP-related SNPs used in our study, 2 SNPs (rs2794520 and rs1260326) were consistent, and 3 correlated SNPs (rs12133641, rs79320731, and rs429358) were included in the same LD blocks. In addition, their effecting alleles showed a consistent directionality of effect of effective allele in terms of their impact on CRP levels was consistent. However, EAF (the effective allele frequency (EAF) or effect size of selected SNPs were different, which is presumed to be due to ethnic differences. The EAF and effect size of rs1260326 were 0.450 and 0.008, respectively, in our study, whereas compared to 0.61 and -0.050, respectively, in the study of Hua et al. In particular, rs1880241, which was included in the study of Hua et al. was excluded from our CRP GWAS because of its low MAF in the East Asian population (<0.01). Nonetheless, we could not find a causal effect of CRP

levels on survival ~~of in~~ CRC patients in this East Asian population, ~~as well as in~~ similar to findings in the previously studied European population.

Although not MR studies, ~~there are~~ several previous genetic studies ~~for have investigated~~ survival in CRC patients. Two ~~previous studies have~~ evaluated associations between CRP-related SNPs and mortality in CRC patients, but the results were also not significant [21,22]. Two GWASs ~~have~~ examined survival in CRC patients. In the Scottish Colorectal Cancer Study [46], no variants reached ~~a~~ the P -value threshold for statistical significance. In ~~comparison contrast~~, Phipps et al. reported that rs209489 was associated with poor survival in patients with distant metastatic CRC [47]. However, since rs209489 was not significantly associated with ~~the~~ log₂-transformed CRP levels in our GWAS, we did not consider this SNP.

Regarding the association between CRP levels and CRC survival, ~~in contrast contrary~~ to our findings, previous observational studies have reported that ~~a~~ high CRP levels ~~was were~~ associated with a poor CRC prognosis. A meta-analysis of 21 observational studies reported that an elevated preoperative CRP level was associated with poor survival with pooled HRs (~~95% CI~~) of 2.04 (95% CI, 1.45–2.85) for overall survival and 4.37 (95% CI, 2.63–7.27) for CRC-specific survival [12]. In a study of CRC patients treated with neoadjuvant therapy and surgery, CRP ~~was levels were~~ associated with disease-free survival independently of carcinoembryonic antigen (~~CEA~~) levels or resection margins [5]. Another meta-analysis using the CRP-to-albumin ratio had similar results [13]. However, since the treatment, stage, and CRP cutoff varied in those studies, it is difficult to distinguish between the effects of cancer-related inflammation and circulating CRP. The prognostic role of CRP levels in CRC patients remains ~~controversial a matter of debate~~ in studies of cancer-free general populations. ~~The In the general population, the~~ effects of cancer-related inflammation on the association between CRP and CRC prognosis would be reduced, ~~although inconsistent findings have been reported regarding a the~~ positive association between elevated pre-diagnostic CRP levels and CRC mortality ~~in the general population remains controversial~~. In the National Health and Nutrition Examination Survey (~~NHANES~~) III, ~~the~~ CRP levels ~~was were~~ positively associated with ~~the~~ CRC-specific mortality in the general population [3], while the Apolipoprotein Mortality Risk Study (~~AMORIS~~) reported a null association [48], and the

266 Copenhagen City Heart Study [49] reported a possible association between baseline CRP levels and
267 CRC-specific mortality in their cohort.

268 Chronic inflammation induces cancer invasion, progression, and metastasis, and influences the
269 efficacy of chemotherapy and immunotherapy [50]. CRP, ~~which is~~ synthesized in the liver, ~~and~~ is an
270 acute-phase protein that reflects inflammation. However, we did not find an effect of genetically
271 predicted CRP levels on CRC survival as a systemic inflammatory mediator, unlike many previous
272 case-control studies [5,7,15], ~~and~~ prospective studies [3,4], and meta-analyses [10,12,13]. However,
273 since the potential causal role of CRP was not confirmed in previous Mendelian studies, the observed
274 effect of CRP elevation on CRC mortality may be due to residual effects or reverse causality.

275 Our study had several limitations. First, since information on recurrence was not available,
276 disease-free survival was not included in the analysis. Second, the ~~batch-effect may not have been~~
277 ~~excluded in CRP and CRC GWAS. Therefore, to minimize the batch-effect, we evaluated the~~
278 ~~association between genetic variants and serum CRP levels by statistically adjusting for survey years~~
279 ~~and study sites in the CRP GWAS, and the association between genetic variants and survival by~~
280 ~~statistically adjusting for the genotyping arrays in the CRC GWAS. Third, because there were only 804~~
281 ~~metastatic CRC cases were only 804 in our study, a further evaluation is needed for to clarify the effect~~
282 ~~of CRP levels on survival in patients with metastatic CRC-survival. Fourth, to confirm the null~~
283 ~~association between CRP levels and mortality in CRC patients, future studies with higher statistical~~
284 ~~power are needed. In particular, to improve the power of the MR analysis, the CRP variance explained~~
285 ~~by the genetic instruments should be discussed in terms of to discuss the biological effects of selected~~
286 ~~SNPs or CRP levels on cancer mortality to compensate for, because of the lack of functional analysis~~
287 ~~of SNPs.~~

288 In summary, we found that ~~the~~ genetically predicted CRP levels ~~was~~ ~~were~~ not associated with
289 the overall or CRC-specific survival of CRC patients. Therefore, our results suggest that genetically
290 predisposed circulating CRP levels do not play a causal role in the prognosis of CRC.

291

메모 포함[오전4]: In the original part of the sentence, this independent clause was missing a main verb, making the precise implications somewhat difficult to interpret. Please confirm that this revision accurately clarifies your intended meaning.

292 **Ethics Statement**

293 The KoGES was reviewed and approved by the Korea Centers for Disease Control and
294 Prevention in Korea (IRB No. 2015-08EXP-01-C-A and 2016-02-20-C-A). Informed consent was
295 obtained from the participants.

296 The HCES-CRC was reviewed and approved by the CNUHH IRB (IRB No. CNUHH-2020-
297 063). All patients and controls gave informed consent to study participation at the time of peripheral
298 blood collection.

299 **Conflict of Interest**

300 The authors have no conflict of interest to declare for this study.

302 **Author Contributions**

303 Conceptualization: SSK. Data curation: SSK, MHS. Formal analysis: CKC, YHY. Funding
304 acquisition: SSK. Investigation: SSK, MHS, SHC. Methodology: SSK, CKC. Writing -
305 original draft: SSK, CKC. Writing - review & editing: SSK, CKC, YHY, MHS, SHC.

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Table 1. Selected SNPs ~~using~~ used as instrumental variables for log₂-transformed CRP levels

SNP	Chr	Nearby gene	Position*	EA	RA	CRP GWAS ^a				CRC GWAS ^b				
						EAF	F	Beta	P-value	Overall mortality		CRC mortality		P-value
										HD	(/ 1,000 PY)	HD	(/ 1,000 PY)	
rs2794520	1	<u>CRP</u>	159678816	C	T	0.365	961.9	0.247	7.23x10⁻²⁰⁰	0.366	-0.230	0.887	0.018	0.989
rs12133641	1	<u>IL6R</u>	154428283	G	A	0.433	196.99	-0.116	3.52x10⁻⁴⁷	0.429	-1.860	0.210	-0.454	0.701
rs71086917	1	<u>LINC02819</u>	159455503	insA	-	0.452	61.2	-0.064	6.54x10⁻¹⁶	0.414	0.127	0.938	0.934	0.493
rs1260326	2	<u>GCKR</u>	27730940	C	T	0.450	133.0	-0.090	1.31x10⁻²⁹	0.452	3.017	0.050	1.535	0.201
rs7383869	7	<u>IL6</u>	22748190	A	G	0.362	164.1	0.105	3.53x10⁻³⁷	0.365	0.571	0.693	0.562	0.620
rs79320731	12	<u>HNF1A</u>	121422449	CTGACTGGCACTCAGCA	T	0.455	461.1	-0.169	1.22x10⁻⁹⁹	0.454	-0.929	0.530	-1.999	0.095
rs429358	19	<u>APOE</u>	45411941	C	T	0.096	587.6	-0.321	3.53x10⁻¹²⁵	0.057	8.684	0.011	7.331	0.014

^a Age, sex, study centers, survey years, and first 10 principal components were adjusted in all models.

^b Age, sex, genotyping array, TNM stage, and first 10 principal components were adjusted in all models.

*~~Position~~ The positions of the SNPs were derived from GRCh37.

CRC: colorectal cancer; CRP: C-reactive protein; EA: effective allele; EAF: effective allele frequency; GWAS: genome-wide association study; HD: hazard difference; PY: person-year; RA: reference allele; SNP: single-nucleotide polymorphism.

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Table 2. Mendelian randomization results for the effects of serum CRP levels on overall and CRC-specific mortality

	Main analysis (Seven -7 serum CRP <u>level</u> -related SNPs)				Sensitivity analysis (ex -cluding rs429358)			
	Overall mortality HD (/ 1,000 PY, 95% CI)	<u>P</u> - value	CRC-specific mortality HD (/ 1,000 PY, 95% CI)	<u>P</u> - value	Overall mortality HD (/ 1,000 PY, 95% CI)	<u>P</u> - value	CRC-specific mortality HD (/ 1,000 PY, (95% CI)	<u>P</u> - value
Using summary statistics ^a								
IVW	-2.92 (-14.05, 8.21)	0.607	-0.76 (-9.61, 8.08)	0.866	1.11 (-8.18, 10.40)	0.815	2.40 (-4.51, 9.31)	0.497
Simple median	-0.93 (-13.67, 11.81)	0.886	0.07 (-10.64, 10.78)	0.989	2.25 (-9.74, 14.24)	0.713	2.00 (-7.80, 11.80)	0.689
Weighted median	1.43 (-8.96, 11.81)	0.788	1.70 (-7.00, 10.41)	0.701	3.39 (-7.00, 13.78)	0.523	2.71 (-5.97, 11.39)	0.540
MR-Egger intercept	0.92 (-3.29, 5.12)	0.669	0.27 (-3.11, 3.65)	0.876	-0.57 (-4.21, 3.08)	0.760	-1.02 (-3.52, 1.49)	0.426
Using individual GRS ^b								
GRS per 1 SD increment	-3.18 (-5.77, -0.59)	0.016	-2.09 (-4.26, 0.08)	0.056	0.17 (-1.82, 2.16)	0.870	0.11 (-1.49, 1.70)	0.897
GRS quintile								
1st	8.53 (1.12, 15.94)	0.024	5.51 (-0.71, 11.73)	0.082	1.94 (-4.47, 8.34)	0.554	2.39 (-2.17, 6.95)	0.349
2nd	1.86 (-5.16, 8.88)	0.603	1.04 (-4.85, 6.93)	0.730	-2.15 (-8.56, 4.27)	0.512	-1.41 (-5.96, 3.14)	0.598
3rd	1 (reference)		1 (reference)		1 (reference)		1 (reference)	
4th	-2.64 (-9.13, 3.85)	0.425	-1.54 (-6.68, 3.60)	0.557	-0.52 (-6.74, 5.70)	0.870	-2.06 (-6.53, 2.40)	0.417
5th	1.14 (-5.17, 7.45)	0.723	0.99 (-4.12, 6.09)	0.705	2.00 (-4.45, 8.45)	0.554	3.16 (-1.45, 7.77)	0.229

^a Results are expressed per ~~two~~2-fold increase in serum CRP levels.

^b Age, sex, genotyping array, and TNM stage were adjusted.

CRC= colorectal cancer; CRP= C-reactive protein; GRS= genetic risk score; HD= hazard difference; PY= person-year; IVW= inverse-variance weighted method; SD= standard deviation; SNP= single-nucleotide polymorphism.

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Table 3. Subgroup analysis by metastasis of CRC

	Non-metastatic CRC				Metastatic CRC			
	Overall mortality HD (/ 1,000 PY, 95% CI)	P value	CRC-specific mortality HD (/ 1,000 PY, 95% CI)	P value	Overall mortality HD (/ 1,000 PY, 95% CI)	P value	CRC-specific mortality HD (/ 1,000 PY, 95% CI)	P value
Using summary statistics ^a								
IVW	1.73 (-6.95, 10.41)	0.697	3.37 (-3.41, 10.15)	0.330	-27.69 (-95.81, 40.43)	0.426	-28.19 (-91.79, 35.42)	0.385
Simple median	3.57 (-6.96, 14.09)	0.506	2.96 (-5.93, 11.85)	0.514	-24.08 (-121.25, 73.10)	0.627	8.90 (-85.99, 103.79)	0.854
Weighted median	4.73 (-3.69, 13.16)	0.271	5.11 (-1.29, 11.51)	0.117	-42.26 (-125.36, 40.83)	0.319	-30.04 (-108.51, 48.43)	0.453
MR-Egger intercept	0.04 (-3.37, 3.44)	0.984	-0.68 (-3.24, 1.89)	0.605	12.62 (-10.94, 36.18)	0.294	12.53 (-9.56, 34.62)	0.266
Using individual GRS ^b								
GRS per 1 SD increment	-1.74 (-3.95, 0.46)	0.121	-0.45 (-2.20, 1.30)	0.613	-18.65 (-42.48, 5.17)	0.125	-22.65 (-45.57, 0.27)	0.053
GRS quintile								
1st	3.00 (-3.48, 9.49)	0.364	0.37 (-4.51, 5.26)	0.881	61.88 (-5.52, 129.29)	0.072	60.52 (-4.77, 125.81)	0.069
2nd	-2.96 (-8.76, 2.84)	0.317	-3.08 (-7.39, 1.22)	0.160	45.84 (-14.95, 106.64)	0.139	37.64 (-21.03, 96.31)	0.209
3rd	1 (reference)		1 (reference)		1 (reference)		1 (reference)	
4th	-0.76 (-6.00, 4.49)	0.778	0.61 (-3.21, 4.43)	0.755	-31.00 (-80.02, 18.02)	0.215	-30.44 (-72.34, 11.46)	0.155
5th	-0.87 (-6.38, 4.65)	0.758	-0.07 (-3.94, 3.80)	0.972	17.10 (-35.25, 69.44)	0.522	2.43 (-43.66, 48.53)	0.918

^a Results are expressed per ~~two~~2-fold increase in serum CRP levels.

^b Age, sex, and genotyping array were adjusted.

CRC: colorectal cancer; CRP: C-reactive protein; GRS: genetic risk score; HD: hazard difference; PY: person-year; IVW: inverse-variance weighted method; SD: standard deviation.

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